

Kunj Rathod

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Education

University of Utah

B.S. in Computer Science

Aug 2023 – Dec 2026

GPA: 3.7/4.0 — Dean's List

Coursework: Computer Vision, NLP, Machine Learning, Distributed Systems, Operating Systems, Algorithms

Research

Spatial Memory for Embodied Agents & Long-Horizon Web Agents

Jan 2026 – Present

University of Utah

- Designing retrieval-augmented spatial memory architectures for embodied agents, unifying spatial, temporal, episodic, and semantic memory for long-horizon robot navigation and manipulation.
- Evaluating agent benchmarks (OS Marathon, BrowseComp, Mind2Web) across live/offline settings; investigating automatic simulation environment generation from natural language (SceneSmith) for scalable zero-shot robot policy evaluation.

Circuit-Level Generalization in LLMs via Sparse Feature Circuits

Jan 2026 – Present

University of Utah

- Building automated pipelines to extract sparse feature circuits from default tasks and predict generalization to non-default tasks using attribution graphs and Pathways Discovery; analyzing circuit stability across model families.

Multi-Agent Materials Discovery for Extreme Environments

Aug 2025 – Feb 2026

STARS Lab, University of Utah (Collaboration: NASA, Microsoft, U.S. DoD)

- Built multi-agent, graph-augmented extraction pipeline normalizing material-property data from 1,000+ papers into a physics-aware knowledge graph; constraint-based design region engine enforcing temperature, creep, and pressure limits for aerospace material selection with benchmarked extraction accuracy and plot fidelity.

Industry Experience

Software Engineering Intern — Azure Data

May 2026 – Aug 2026

Microsoft, Redmond, WA

- Incoming: building scalable distributed data infrastructure within the Azure ecosystem.

Software Development Intern, AI Services (SUDO Program)

Jan 2025 – Present

University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: token-streaming inference achieving p95 <200ms TTFT, 40% latency reduction via pipeline optimization and API caching; distributed session persistence (DynamoDB + S3) for 1,000+ concurrent conversations.
- Shipped 6 features across 4 sprints: Bedrock Agents, Knowledge Bases, Guardrails; owned API design, schema, and AWS infrastructure (Lambda, API Gateway, RDS).

AI Engineering Intern

Nov 2024 – Apr 2025

CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) to 10M+ documents; +28% recall, -35% hallucinations. Benchmarked 8 LLM families on F1/latency/cost across LegalBench and NyayaAnumana; model routing cut inference spend.

Projects

HirePilot — Autonomous AI Recruiting Pipeline

Mar 2026

Podium AI Hackathon Winner Python, Claude API, Twilio, Perplexity API, GitHub API

- Built 9-agent pipeline replacing end-to-end recruiting: Slack-triggered GitHub sourcing, Perplexity-enriched profiles, live adaptive voice interviews via Twilio + Claude, multi-dimensional scoring with ranked shortlist delivery. Shipped in 24 hours.

Wingman.ai — Multi-Modal iOS AI Assistant SwiftUI, GPT-4.1, Whisper, Firebase

Apr 2025 – Present

- iOS assistant with vision/speech/chat, custom audio streaming with queuing and cancellation, offline-first with Firebase sync.

BioGraphRAG — Distributed Biomedical Knowledge Graph System

May 2024 – Jan 2025

Python, NebulaGraph, LlamaIndex, Docker, AWS

- Engineered GraphRAG system over 1M+ biomedical entities (UniProt, AlphaFold, RXNav); 40% accuracy gain over embedding-only baselines. Optimized graph traversal 3x via caching and node pruning (p95 <500ms).

FlowVia — V2X Edge Traffic Optimization

Apr 2024

Python, TensorFlow (LSTM), V2X (DSRC/C-V2X), OBD-II

- Designed real-time speed recommendation from live SPaT sensor data; LSTM traffic prediction retrained on streaming historical data; AES-256 encrypted V2V/V2I communication with edge-first processing for sub-10ms latency.

Technical Skills

Languages: Python, C++, Java, Swift, TypeScript, SQL

AI/ML: PyTorch, TensorFlow, RAG/GraphRAG, LangChain, LlamaIndex, CrewAI, Multi-Agent Systems, Sparse Circuits, Embodied AI

Systems & Cloud: AWS (Lambda, Bedrock, S3, DynamoDB, RDS), Docker, Kubernetes, CI/CD, FastAPI, gRPC, Edge Deployment

Robotics & Perception: Spatial Memory, Scene Generation, V2X Protocols, Real-Time Sensor Fusion, Simulation Environments